SAMI BIN AYYAD GROUP · عيادبن سامي مجموعة

Environmental Solutions & Sustainability Profile

Comprehensive Environmental Engineering, Waste-to-Energy & Carbon Management

Environmental Solutions

Key Document Highlights & Verified Scope

- 19 Specialized Environmental Consultancy & Engineering Services
- Strategic Alliance with RENERGON International AG (Switzerland)
- Proprietary RSD Dry Fermentation Biogas & Waste-to-Energy Technology
- Scopes 1, 2, 3 Carbon Footprinting, CBAM Compliance & ESG Reporting
- Proprietary Bioremediation (ET-POT-01) & Odor Control (ET-POR-01) Formulations

Official Verified Operational Benchmarks

1977	2023	4	39
Foundational Heritage	Unified Corporate Group	Integrated Sectors	Specialized Services

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Official Corporate Documentation for International Procurement, Investors & B2B Commercial Partners

2026 Environmental Technology Edition | Doc Ref: SBA-ENV-SOL-2026

1. Sector Positioning & Scientific Foundation

European Technology Alliance and Environmental Engineering Leadership

PATENTED RSD DRY ANAEROBIC DIGESTION TECHNOLOGY

Strategic Alliance with RENERGON International AG (Switzerland)

Leading the waste-to-energy and biogas transformation in Egypt and Africa.

Sami Bin Ayyad for Sustainable Environmental Services operates in formal professional alliance with RENERGON International AG. Utilizing RENERGON's patented Simultaneous Digestion (RSD) dry fermentation technology, we design, build, and operate turnkey plants converting municipal solid waste (MSW), agricultural residues, food waste, and manure into clean electricity, biogas, biofuel, biochar, and premium organic compost.

- Processes unsegregated high-solids organic waste (up to 60 % of municipal streams)
- Zero process wastewater discharge with closed-loop thermal efficiency
- High methane yield: up to 120–160 m³ biogas per tonne of organic waste
- Turnkey EPC delivery for small, medium, and utility-scale installations

2. Six Environmental Service Categories (19 Disciplines)

Comprehensive Scientific, Advisory, and Industrial Compliance Solutions

Category	Primary Services Included	Target Industries	Key Standards & Frameworks
1. Consultancy & Compliance	EIA, ESIA, SEA, Permitting, Auditing	Industry, Infrastructure, Energy	EEAA Law 4/1994, IFC, World Bank
2. Waste & Circular Economy	Solid Waste, Hazardous Waste, Recycling	Manufacturing, Municipalities, Petrochem	Egyptian Waste Law 202/2020
3. Renewable Energy & WtE	RENERGON Biogas, WtE Plants, Solar PV	Agri-Business, Heavy Industry, Utilities	Patented RSD Technology
4. Water & Wastewater	Industrial Effluent Treatment, Bioremediation	Food Processing, Oil & Gas, Chemicals	Law 93/1962 & Law 48/1982
5. Carbon Management	Scopes 1-2-3 Auditing, CBAM, ESG, SDGs	Exporters, Multinational Corps, Finance	GHG Protocol / ISO 14064 / GRI
6. Tech & Digital Solutions	IoT Sensing, EHS Platforms, Applied R&D	Smart Cities, Industrial Parks, Ports	Integrated QHSE Cloud Architecture

3. Proprietary Environmental Technologies & Products

Field-Tested Reagents and Real-Time Environmental Measurement Systems

ADVANCED MICROBIAL CONSORTIUM

ET-POT-01: Oil Terminator Bioremediation Agent

Concentrated bacterial formulation rapidly digesting petroleum hydrocarbons and oil spills.

Blended from 8 non-pathogenic bacterial strains (Pseudomonas, Bacillus, Rhodococcus isolates) with count ≥ 5 × 10^9 CFU/g. Achieves up to 85% Total Petroleum Hydrocarbon (TPH) reduction within 30 days without harmful synthetic surfactants.

- Remediates crude oil and fuel spills on land, sand, and coastal environments
- Biologically degreases industrial storage tanks and wastewater treatment lagoons
- Certified Class 1 non-pathogenic, non-toxic to marine and terrestrial ecosystems

CATALYTIC BIOCHEMICAL BOND CLEAVAGE

ET-POR-01: Industrial Odor Neutralizer Solution

Rapidly eliminates hydrogen sulfide (H S), mercaptans, and ammonia at industrial emission sources.

Permanent biochemical breakdown of volatile organic sulfur and nitrogen compounds rather than temporary masking. Applied in wastewater lift stations, composting facilities, landfills, and rendering plants.

Corporate Contact & Business Inquiry

Direct communication desk for partners, clients, and institutional inquiries

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